

DATA ANALYTICS

01

Introduction to Data Analytics

See data the way businesses do — and know exactly which tool to reach for.

DURATION	LEVEL	MODULE	UPDATED
6 Sessions	Beginner	Module 1	June 2026

IN THIS MODULE

01	What is Data Analytics? Overview of Industry Tools
02	Data & Databases — Structured vs Unstructured
03	Domain Understanding for Analytics
04	The Data Analytics Workflow
05	Exploratory Data Analysis (EDA) Overview
06	Case Study Orientation

SESSION

01

60 MIN · FUNDAMENTALS

What is Data Analytics? Overview of Industry Tools

◆ Learning Objectives

- Distinguish between Data, Information, and Insights with real examples
- Identify the four types of analytics: Descriptive, Diagnostic, Predictive, and Prescriptive
- Map the key industry roles: DA, BA, DE, and DS — and how they differ
- Understand where Excel, SQL, Python, and Power BI each fit in the analytics stack
- Observe the same dataset being explored across all four major tools

01 Data -> Information -> Insights

Raw data on its own means nothing. A list of 10,000 numbers from a Flipkart warehouse is just noise. When you calculate that returns spiked 40% on Mondays, that becomes information. When you recommend reducing same-day delivery on weekends to cut return costs, that becomes an insight — the thing businesses actually pay for.

CONCEPT — The Data-Insight Chain

Data is raw facts (sales figures, click counts). Information is data with context (sales fell 12% in Q3). Insight is the 'so what' — the actionable conclusion a manager can act on. A data analyst's job is to convert data into insights as efficiently as possible.

02 Four Types of Analytics

Type	Question It Answers	Example
Descriptive	What happened?	Monthly sales report for a Gujarat textile firm
Diagnostic	Why did it happen?	Why did revenue drop in Surat stores in October?
Predictive	What will happen?	Which customers are likely to churn next month?
Prescriptive	What should we do?	Offer a 10% discount to high-churn-risk customers

PRO TIP — Know Your Deliverable

Most entry-level DA jobs involve Descriptive and Diagnostic analytics. Predictive and Prescriptive require ML skills. Knowing which type your manager expects prevents you from over-engineering a simple dashboard.

03 Industry Roles: Who Does What?

Data Analyst (DA)

Cleans data, builds reports, answers business questions. Primary tools: Excel, SQL, Power BI. Most common entry role.

Data Scientist (DS)

Builds predictive models, applies ML. Primary tools: Python, R, cloud ML platforms. Requires stronger maths background.

A Business Analyst (BA) focuses on translating business requirements into data problems — they often sit between the business team and the DA. A Data Engineer (DE) builds the pipelines that move and store data, so the analyst always has clean, fresh data to work with.

04 The Analytics Tool Stack

Every analyst uses a combination of tools — not just one. Think of them as layers: Excel for quick exploration, SQL for querying databases, Python for complex transformations and analysis, and Power BI for presenting results to stakeholders.

Tool	Primary Use	When You Reach For It
Excel	Quick exploration, pivot tables, dashboards	Small-to-medium datasets, non-technical audiences
SQL	Query large databases	Whenever data lives in a database (almost always)
Python	Cleaning, analysis, automation, ML	Large data, automation, statistical work
Power BI	Interactive dashboards & reports	Presenting insights to business stakeholders

TRY THIS LIVE

Open any simple dataset (even your own phone's screen-time export). Identify: one descriptive fact, one diagnostic question you could ask, and one prediction you could make. Share with the batch.

CASE STUDY — Jio Platforms & Data-Driven Decisions

Context — Jio Platforms, India's largest telecom operator, processes data from 450+ million subscribers across Gujarat and the rest of India.

Challenge — With millions of daily recharges, Jio needed to understand why certain recharge plan sizes were declining in Tier 2 cities like Rajkot and Vadodara.

What they did — Their data analysts used descriptive analytics (recharge trends by city) and diagnostic analytics (correlating plan size with network speed and competitor pricing) to identify the root cause.

Result — Targeted plan bundles were launched for Tier 2 markets, recovering 8% of lapsed subscribers within one quarter.

Takeaway — The four analytics types are not academic — they map directly to real business decisions companies make every week.

Source: Jio Annual Report 2023; public press coverage

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Introduction to Data Analytics — Lecture 1: Overview and Motivation (IIT Roorkee)

Link — <https://nptel.ac.in/courses/110107092>

Always link the official channel — beware fan-made or re-upload copies.

“The goal is to turn data into information, and information into insight. — Carly Fiorina, former CEO, HP”

WHERE TO GO NEXT

You now know what analytics is and which tools the industry uses. Next, we go deeper into where data actually lives.

Explore deeper — Google 'DA vs DS vs DE salary India 2024' — compare the roles you learned today with real market data.

Try on your own — Find one public dataset on data.gov.in and identify which type of analytics question you could answer with it.

Connect the dots — Session 2 covers data formats and databases — where your raw data comes from before any analysis begins.

Going further — Read 'Competing on Analytics' by Davenport & Harris — the foundational book on building data-driven organisations.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analytics trainer. Explain the difference between Descriptive, Diagnostic, Predictive, and Prescriptive analytics using examples from an Indian e-commerce company like Meesho or Flipkart. Keep it under 150 words and use simple Hinglish.

SESSION

02

60 MIN · FUNDAMENTALS

Data & Databases — Structured vs Unstructured

Learning Objectives

- Identify the most common data formats: CSV, XLSX, JSON, SQL, and APIs
- Distinguish between structured and unstructured data with real examples
- Understand what an RDBMS and a Data Warehouse are and how they differ
- Explain the difference between ETL and ELT pipelines
- Differentiate OLTP and OLAP systems and when each is used

01 Data Formats You Will Encounter Every Day

Before you can analyse anything, you need to know what shape your data arrives in. A grocery chain in Ahmedabad might send you daily sales as a CSV file, their customer profiles as JSON from an API, and their transaction history stored in a MySQL database — all for the same project.

Format	What It Looks Like	Common Source
CSV	Comma-separated rows of values — opens in Excel	Exports from ERP systems, Kaggle datasets
XLSX	Excel workbook with multiple sheets and formatting	Finance teams, operations reports
JSON	Key-value pairs, nested structure — human-readable	APIs, web apps, NoSQL databases
SQL Table	Rows and columns inside a relational database	CRM, banking systems, e-commerce backends
API	Live data stream accessed via a URL request	Zomato, Razorpay, government open data portals

CONCEPT — Structured vs Unstructured Data

Structured data fits neatly into rows and columns (a sales table, a customer list). Unstructured data has no fixed format — think customer reviews, WhatsApp messages, call recordings, or social media posts. Around 80% of enterprise data is unstructured. As a DA, you'll mostly work with structured data, but you need to know that unstructured data exists and that tools like NLP are used to analyse it.

02 Databases: RDBMS and Data Warehouses

An RDBMS (Relational Database Management System) stores structured data in related tables. Every bank branch in India, every hospital, and every e-commerce company uses one. MySQL, PostgreSQL, and Oracle are the most common in the Indian market.

A Data Warehouse is a separate, large-scale storage system built specifically for analytics — not daily operations. Companies like D-Mart or Tata Retail pull data from dozens of operational databases into a central Data Warehouse so analysts can run heavy queries without slowing down the live business systems.

RDBMS (Operational DB)

Handles day-to-day transactions. Fast reads/writes on small data. Examples: MySQL, PostgreSQL. Used by billing systems, CRMs.

Data Warehouse (Analytical DB)

Stores historical data for analysis. Handles huge queries across years of data. Examples: Amazon Redshift, Google BigQuery, Snowflake.

03 ETL vs ELT — Moving Data Into the Warehouse

CONCEPT — ETL vs ELT

ETL (Extract -> Transform -> Load): Data is cleaned and transformed before it enters the warehouse. Traditional approach, used when warehouses had limited compute. ELT (Extract -> Load -> Transform): Raw data lands in the warehouse first, then is transformed there using the warehouse's own compute power. Modern cloud warehouses like BigQuery prefer ELT because storage is cheap and compute is powerful.

04 OLTP vs OLAP — Two Very Different Jobs

Property	OLTP	OLAP
Full form	Online Transaction Processing	Online Analytical Processing
Purpose	Record daily transactions	Analyse historical data
Query type	Simple: INSERT, UPDATE, SELECT one row	Complex: aggregations across millions of rows
Example	A Swiggy order being placed and recorded	Monthly revenue by city for the last 3 years
System	MySQL, Oracle DB	Amazon Redshift, Power BI datasets

TRY THIS LIVE

Open the same CSV dataset in Excel, then paste a few rows into a free online SQL editor (like sqliteonline.com). Notice how the same data looks different across tools. Which felt easier to filter? Which gave you more control?

CASE STUDY — HDFC Bank's Data Infrastructure

Context — HDFC Bank, one of India's largest private banks with major operations across Gujarat, processes over 10 million transactions daily.

Challenge — Analysts needed to run customer behaviour reports, but running heavy queries on the live transaction database was slowing down ATM and net-banking response times.

What they did — HDFC separated their OLTP system (live transactions) from an OLAP data warehouse where analysts could run queries freely. ETL jobs ran overnight to load the previous day's data into the warehouse.

Result — Analyst query times dropped from hours to minutes, and zero impact on live banking performance was reported.

Takeaway — The OLTP/OLAP separation is not just theory — it is the architecture behind every major bank, retailer, and platform in India.

Source: HDFC Bank Technology Annual Report 2022; ET BFSI coverage

“In God we trust; all others must bring data. — W. Edwards Deming”

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Introduction to Data Analytics — Lecture 2: Data Sources and Storage (IIT Roorkee)

Link — <https://nptel.ac.in/courses/110107092>

WHERE TO GO NEXT

You now know where data comes from and how it is stored. The next step is understanding the business context around it.

Explore deeper — Search 'OLTP vs OLAP GeeksforGeeks' for a clean diagram that shows the architecture side by side.

Try on your own — Download any dataset from data.gov.in and identify: Is it structured or unstructured? What format is it in? What system likely generated it?

Connect the dots — Session 3 introduces domain knowledge — because knowing the data format is not

enough if you don't understand the business it came from.

Going further — Read about the Medallion Architecture (Bronze -> Silver -> Gold layers) used in modern data lakehouses like Databricks — the next evolution of data warehousing.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a database teacher. Explain the difference between OLTP and OLAP using an Indian retail store example — like a Reliance Smart or D-Mart store. Include what kind of queries each system handles. Keep it under 150 words in simple Hinglish.

SESSION

03

60 MIN · FUNDAMENTALS

Domain Understanding for Analytics

◆ Learning Objectives

- Explain why domain knowledge is as important as technical skills for a data analyst
- Identify key business domains and the questions analysts answer in each
- Define KPIs and business metrics for a given domain
- Pick a domain (Retail) and map it to specific, measurable KPIs
- Understand how the same data can mean different things in different industries

01 Why Domain Knowledge Makes or Breaks an Analyst

Two analysts can have identical Python and SQL skills. The one who understands the business will always produce better insights. A churn rate of 5% means nothing unless you know that in the telecom industry, anything above 2% monthly is a crisis — while in SaaS, 5% annually is considered healthy.

CONCEPT — Domain Knowledge

Domain knowledge is your understanding of how a specific industry works — its goals, processes, vocabulary, and what 'good' looks like in terms of numbers. Without it, you can produce technically correct charts that lead to completely wrong decisions. A DA who understands retail will notice that comparing sales figures without accounting for Diwali seasonality is meaningless — but only if they know the business.

ANALYSTS WHO LACK DOMAIN CONTEXT

67%

Source: Gartner Data & Analytics Survey, 2022 — of analytics projects that fail to deliver business value cite lack of business context as the primary cause

02 Key Business Domains & Their Analytics Questions

Domain	Core Business Goal	Typical Analytics Question
Retail	Sell more, waste less	Which products have the highest return rate by city?
Finance & Banking	Manage risk, grow loans	Which customer segments are most likely to default?

Domain	Core Business Goal	Typical Analytics Question
HR	Hire and retain talent	What is the average time-to-hire for technical roles?
Healthcare	Improve patient outcomes	Which departments have the longest patient wait times?
E-Commerce	Maximise conversion & repeat purchase	What is the drop-off rate at the payment page?

Notice that each question uses domain-specific vocabulary. An analyst working in HR needs to know what 'attrition', 'headcount', and 'cost-per-hire' mean before they can build a single report. This vocabulary is learned by spending time with the business — not just the data.

03 KPIs and Business Metrics — The Numbers That Matter

CONCEPT — KPI vs Metric

A metric is any measurable number (total orders, page visits, app downloads). A KPI (Key Performance Indicator) is a metric that is tied to a specific business goal and tracked against a target. 'Number of orders' is a metric. 'Number of orders this month vs target of 50,000' is a KPI. Analysts work with both, but KPIs are what leadership cares about.

04 Demo Domain: Retail — Mapping KPIs

Let us take a retail business — imagine a chain of garment stores across Ahmedabad, Surat, and Vadodara. The owner wants to know if the business is healthy. Here are the KPIs a data analyst would define and track.

KPI	Formula	Why It Matters
Revenue	Units Sold × Selling Price	Top-line health of the business
Customer Acquisition Cost (CAC)	Total Marketing Spend ÷ New Customers	Are we spending too much to get each buyer?
Profit Margin	(Revenue - Cost) ÷ Revenue × 100	Is the business actually making money?
Average Order Value (AOV)	Total Revenue ÷ Number of Orders	Are customers buying more per visit over time?
Inventory Turnover	Cost of Goods Sold ÷ Avg Inventory	Are products sitting too long on the shelf?

TRY THIS LIVE

Pick any one domain from the table above (Retail, Finance, HR, Healthcare, E-Commerce). Write down three KPIs you think that domain tracks. Then Google '[domain] KPIs list' and compare your list to industry-standard ones. How close were you?

CASE STUDY — Amul's Data-Driven Supply Chain

Context — Amul, headquartered in Anand, Gujarat, is one of India's largest dairy cooperatives — managing milk collection from 3.6 million farmers across Gujarat.

Challenge — Amul needed to track procurement, processing, and distribution KPIs across thousands of village societies to reduce spoilage and ensure supply matched demand by city.

What they did — Domain experts (dairy supply chain specialists) defined KPIs like 'milk procurement per litre cost', 'spoilage rate per district', and 'city-wise demand coverage ratio'. Analysts then built dashboards around these domain-specific metrics.

Result — Spoilage was reduced by 18% and city-level stock-outs dropped significantly, directly improving farmer incomes and product availability.

Takeaway — Without domain experts defining the right KPIs, even the best analyst would have built dashboards tracking the wrong numbers.

Source: Amul Annual Report 2022-23; IIM Ahmedabad case study on Amul's digital transformation

“Data without domain is noise. Domain without data is intuition. You need both.”

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Introduction to Data Analytics — Lecture 3: Business Context and KPIs (IIT Roorkee)

Link — <https://nptel.ac.in/courses/110107092>

WHERE TO GO NEXT

You now know what domain knowledge means and how to define KPIs. The next session connects this to a structured analytics workflow.

Explore deeper — Search 'retail analytics KPIs list 2024' and 'banking analytics KPIs India' — see how companies in your target industry measure performance.

Try on your own — Choose a local Gujarat business you know (a kirana store, a textile shop, a coaching institute). Write five KPIs you would track if you were their data analyst.

Connect the dots — Session 4 shows how these KPIs are actually produced — through the end-to-end data

analytics workflow from requirement to presentation.

Going further — Read 'Measure What Matters' by John Doerr — the definitive book on OKRs and how large organisations define and track business goals.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a business analyst for a mid-sized retail chain in Gujarat. List 5 KPIs the business should track and explain what each one tells the store owner in simple Hinglish. Use realistic Indian numbers as examples. Keep it under 150 words.

SESSION

04

60 MIN · FUNDAMENTALS

The Data Analytics Workflow

◆ Learning Objectives

- Name and sequence all eight stages of the data analytics workflow
- Understand what happens at each stage and what the output is
- Identify common failure points in analytics projects and how to avoid them
- Trace a real business problem through the complete workflow end-to-end
- Know what a data analyst is responsible for vs what other roles handle

01 The 8-Stage Analytics Workflow

Every analytics project — from a simple Excel sales report to a Power BI dashboard for a bank — follows the same fundamental path. Skipping stages is the single biggest reason projects produce wrong or useless outputs. A trainer at a coaching institute who starts building charts before cleaning the data will present insights based on corrupted numbers.

Stage	What You Do	Output
1. Requirement Gathering	Understand the business question — talk to stakeholders	A clear problem statement + list of KPIs to measure
2. Data Collection	Identify sources and pull data (CSV, DB, API, manual)	Raw data files or database tables
3. Data Cleaning	Handle nulls, duplicates, wrong formats, outliers	A clean, trustworthy dataset
4. Data Transformation	Reshape, aggregate, join, create new calculated columns	An analysis-ready dataset
5. Modelling / Analysis	Apply statistics, pivot tables, SQL queries, Python	Numbers, patterns, correlations, trends
6. Reporting	Build tables, charts, dashboards in Excel or Power BI	Visual reports stakeholders can read
7. Insights	Interpret results — explain what the numbers mean	Written or verbal commentary on findings
8. Presentation	Present findings to business stakeholders clearly	Decision-ready recommendations

02 Stage 1 & 2 — Requirements and Data Collection

CONCEPT — The Business Question Comes First

Never start by opening data. Start by asking: 'What decision will this analysis help make?' A manager at a Surat apparel manufacturer may say 'I want to see sales data' — but what they actually need is 'Which product lines should we expand before the wedding season?' These are very different analyses. Always clarify the question first.

03 Stage 3 & 4 — Cleaning and Transformation

Data cleaning is not glamorous, but it is where most analyst time is spent — typically 60–80% of project time. Dirty data means dirty insights. Common issues: blank cells where values should exist, the same city spelled as 'Ahmedabad', 'Ahemdabad', and 'AMD', dates stored as text, and negative values in quantity columns.

WARNING — Garbage In, Garbage Out

If you skip cleaning and jump to analysis, every insight you produce is unreliable. A dashboard built on uncleaned data is worse than no dashboard — because leadership will make real business decisions based on wrong numbers, and they will blame the analyst when outcomes don't match.

04 Stage 5–8 — Analysis, Reporting, Insights & Presentation

Analysis is where you apply statistical techniques, SQL queries, or Python to find patterns. Reporting turns those patterns into visuals. But the most important stage is Insights — explaining what the numbers actually mean in plain language. A chart that shows a 23% revenue dip in Q2 is reporting. Saying 'the dip was caused by a 40-day delay in restocking the top 3 SKUs — here is what we recommend' is an insight.

PRO TIP — Lead With the Insight, Not the Chart

In a business presentation, state the insight in the first sentence. 'Revenue fell 23% in Q2 due to supply chain delays — we recommend a buffer stock policy for top 10 SKUs.' Then show the chart. Most analysts do the opposite and lose the audience before they reach the point.

TRY THIS LIVE

Take this scenario: A coaching institute in Ahmedabad notices enrolments have dropped 15% this quarter. Walk through all 8 stages of the workflow out loud with your batch — What data would you collect? How would you clean it? What would your final insight and recommendation look like?

CASE STUDY — BigBasket's Demand Forecasting Workflow

Context — BigBasket, India's largest online grocery platform with strong operations in Ahmedabad, Surat, and Vadodara, manages 40,000+ SKUs across multiple cities.

Challenge — Stockouts during festival seasons (Navratri, Diwali) were causing order cancellations and customer churn in Gujarat markets.

What they did — Their analytics team followed the full workflow: gathered requirements from supply chain heads (Stage 1), collected historical sales and inventory data (Stage 2), cleaned SKU-level data for duplicates and category mismatches (Stage 3), transformed data into weekly demand patterns by city (Stage 4), modelled seasonal demand spikes (Stage 5), built inventory dashboards (Stage 6), and presented city-specific restocking recommendations to operations (Stage 7–8).

Result — Stockout incidents during the 2022 Navratri season were reduced by 34% compared to the previous year in Gujarat cities.

Takeaway — The workflow is not a checklist — it is a structured thought process. Skipping any stage compounds errors in all the stages that follow.

Source: BigBasket Tech Blog 2022; YourStory coverage of BigBasket's analytics infrastructure

*“An analyst who presents charts without insights is a librarian handing you books.
The insight is the recommendation — that is what you are paid for.”*

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Introduction to Data Analytics — Lecture 4: The Analytics Process (IIT Roorkee)

Link — <https://nptel.ac.in/courses/110107092>

WHERE TO GO NEXT

You now have a complete mental map of how analytics projects are structured. Session 5 zooms into Stage 5 — the exploratory analysis phase — in much more detail.

Explore deeper — Search 'data analyst workflow real project' on Medium or Towards Data Science — read one end-to-end project walkthrough to see the workflow in action.

Try on your own — Draw the 8-stage workflow as a flowchart on paper. Add one common mistake that could happen at each stage. This will be your personal quality checklist for every future project.

Connect the dots — Session 5 covers EDA — the most important part of Stage 5 — where you learn to explore data systematically before drawing any conclusions.

Going further — Read about the CRISP-DM methodology (Cross-Industry Standard Process for Data Mining) — the formal industry standard for analytics project management.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a senior data analyst. Walk me through how you would apply the 8-stage analytics workflow to a real problem: enrolments have dropped at an IT training institute in Gujarat. For each stage, mention what you would do and what tool you would use. Keep it under 200 words in simple English.

SESSION

05

60 MIN · FUNDAMENTALS

Exploratory Data Analysis (EDA) — Overview

◆ Learning Objectives

- Define Exploratory Data Analysis and explain its purpose in a project
- Distinguish between Univariate, Bivariate, and Multivariate analysis
- Identify trends, relationships, and outliers using basic Excel techniques
- Ask the right questions before building any chart or running any formula
- Build a mental framework for approaching any new dataset systematically

01 What Is EDA and Why Does It Come Before Everything Else?

EDA — Exploratory Data Analysis — is the phase where you get to know your data before doing any formal analysis. Think of it like a doctor examining a patient before prescribing treatment. You would never build a sales forecast without first understanding the shape of your data: how many rows, what columns, which values are missing, and whether the numbers even make sense.

CONCEPT — EDA Defined

EDA is an approach of analysing datasets to summarise their main characteristics — often using visual methods — before applying any formal modelling. Coined by statistician John Tukey in 1977, EDA is about generating hypotheses, not testing them. You are not proving anything yet — you are asking: 'What is this data trying to tell me?'

02 Three Types of EDA Analysis

Type	How Many Variables?	What You Are Looking For	Example
Univariate	One variable at a time	Distribution, range, outliers, central tendency	What is the distribution of monthly salaries across 500 employees?
Bivariate	Two variables together	Relationship or correlation between them	Does advertising spend correlate with store revenue?
Multivariate	Three or more variables	How multiple factors interact simultaneously	How do age, city, and income level together affect product preference?

In practice, every EDA starts with Univariate analysis — you understand each column individually first. Then you

move to Bivariate to spot relationships. Multivariate comes last, typically using heatmaps or pair plots in Python (covered in later modules). Today we focus on the first two using Excel.

03 What to Look For in EDA

CONCEPT — Trends, Relationships, and Outliers

A trend is a consistent direction in data over time (sales rising every Diwali). A relationship is when two variables move together (higher discount -> higher quantity sold). An outlier is a data point that is surprisingly far from the rest — it could be an error (someone entered 10,000 instead of 100) or a genuine anomaly (one store in Rajkot sells 5x more than all others — why?).

What to Check	How to Check It in Excel	Red Flag to Watch For
Number of rows & columns	Ctrl+End to jump to last used cell	Far fewer rows than expected — data may be incomplete
Missing values	Filter each column for blanks	More than 5% missing in a key column
Data types	Format Cells — check each column	Dates stored as text, numbers stored as strings
Duplicates	Remove Duplicates tool	Same order ID appearing multiple times
Outliers	Sort column highest to lowest	Values that are 10x the average with no explanation
Basic distribution	Create a histogram or pivot table	Heavily skewed data that may need transformation

TRY THIS LIVE

Open any CSV dataset from Kaggle (try the Superstore Sales dataset). In Excel, do the following: (1) Ctrl+End to see dataset size, (2) filter Sales column for blanks, (3) sort Profit column highest to lowest and look at the top 5 and bottom 5 rows. Write down three observations about what you see — just in plain English.

04 The 5 EDA Questions to Ask Every Time

Before you build a single chart or write a single formula, ask yourself these five questions about any new dataset:

Question	Why It Matters
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Question	Why It Matters
How big is this data? (rows × columns)	Determines which tool to use — Excel struggles beyond 1M rows
What does each column represent?	You cannot analyse what you do not understand
Are there missing or incorrect values?	Null values and data entry errors corrupt every downstream calculation
What is the time range of this data?	Old data may be irrelevant; mixed date ranges cause false trends
What business question am I trying to answer?	Keeps you from exploring endlessly without producing any insight

CASE STUDY — Meesho's Seller Quality EDA

Context — Meesho, Bangalore-based but with a very large seller and buyer base across Gujarat's Tier 2 and Tier 3 cities, onboarded 100,000+ new sellers in 2022.

Challenge — Product return rates spiked after rapid seller onboarding. The analytics team needed to understand which sellers, categories, and geographies were driving the problem.

What they did — The team started with Univariate EDA — distribution of return rates by seller. Then Bivariate — return rate vs product category, and return rate vs seller age (months since joining). They found that sellers less than 3 months old and selling in the Electronics category had a return rate 4x the average.

Result — A new seller onboarding quality gate was added specifically for Electronics, reducing return rates from new sellers by 31% within two quarters.

Takeaway — EDA is not a formality — it is how analysts discover which variables actually matter before building any model or dashboard.

Source: Meesho Engineering Blog 2022; Inc42 coverage of Meesho's seller quality initiatives

“The greatest value of a picture is when it forces us to notice what we never expected to see. — John Tukey, inventor of EDA”

WATCH — VIDEO REFERENCE (HINDI/ENGLISH)

Channel — NPTEL (official — IITs / IISc)

Watch — Introduction to Data Analytics — Lecture 5: Exploratory Analysis Techniques (IIT Roorkee)

Link — <https://nptel.ac.in/courses/110107092>

WHERE TO GO NEXT

You now have a systematic approach for opening any new dataset. Session 6 applies this thinking to the actual case studies you will work on throughout this program.

Explore deeper — Search 'EDA checklist data science' on Towards Data Science — save one checklist you find useful as a reference for your first real project.

Try on your own — Download the Telco Customer Churn dataset from Kaggle. Open it in Excel and answer the five EDA questions from Section 4 for this dataset. Time yourself — can you do it in under 15 minutes?

Connect the dots — Session 6 introduces all the case studies in this program — you will apply the EDA lens to each one before any formal analysis begins.

Going further — Read John Tukey's original 1977 concept summary of EDA — search 'Tukey EDA exploratory data analysis 1977 summary'. It is short and will shift how you think about data permanently.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analyst. I have opened a new dataset with 50,000 rows of retail sales data from Indian stores. Walk me through the first 10 minutes of EDA — what I should check, in what order, and why. Give me specific Excel steps. Keep it under 200 words in simple English.

SESSION

06

60 MIN · FUNDAMENTALS

Case Study Orientation

◆ Learning Objectives

- Understand the structure and purpose of all case studies in the DA program
- Identify the business question, dataset, and expected output for each case study
- Trace how each case study flows through the modules — which tools are applied where
- Apply the EDA lens from Session 5 to preview each dataset
- Set personal learning targets for what you want to build by the end of the program

01 Why Case Studies — Not Just Exercises

Exercises teach you syntax. Case studies teach you thinking. In this program, every tool you learn — Excel, SQL, Python, Power BI — will be applied to the same real-world datasets repeatedly. By Module 10, you will not just know how to use VLOOKUP. You will know when to use it, why, and what insight it helped you produce for a real business problem.

CONCEPT — The Case Study Thread

Each case study in this program is a thread that runs across multiple modules. You will first see the data in Excel (clean and explore it), then query it with SQL, then analyse it with Python, and finally visualise it in Power BI. This repetition across tools is intentional — it builds the muscle memory of thinking analytically, not just technically.

02 Case Study 1 — E-Commerce Sales Analysis

Property	Details
Dataset	Online Retail Dataset (Kaggle) — OnlineRetail.csv
Key Columns	InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID, Country
Business Question	Who are the most valuable customers and what do they buy?
Modules Applied In	Statistics (Module 2), Excel (Module 3)
Expected Output	Customer purchasing behaviour report + t-test on spending patterns

03 Case Study 2 — Customer Churn Analysis

Property	Details
Dataset	Telco Customer Churn Dataset (Kaggle) — Telco-Customer-Churn.csv
Key Columns	CustomerID, Gender, Tenure, InternetService, Contract, MonthlyCharges, TotalCharges, Churn (Yes/No)
Business Question	What factors most strongly predict whether a customer will leave?
Modules Applied In	Statistics (Module 2), Excel (Module 3)
Expected Output	Statistical analysis of churn drivers + retention strategy recommendations

04 Case Study 3 — Retail Store Sales Dashboard

Property	Details
Dataset	Superstore Sales Dataset (Kaggle) — SampleSuperstore.csv
Key Columns	Order Date, Ship Mode, Segment, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit
Business Question	Which regions and product categories are most and least profitable?
Modules Applied In	Excel (Module 3)
Expected Output	Interactive Excel dashboard with regional and category-level profit analysis

05 Case Study 4 — Zomato Bangalore Restaurant Analysis

Property	Details
Dataset	Zomato Bangalore Restaurants — Kaggle
Key Columns	Restaurant name, location, cuisine, rating, votes, approx cost, online order availability

Property	Details
Business Question	Where should Zomato focus marketing and partnerships for maximum growth?
Modules Applied In	SQL (Module 4)
Expected Output	SQL-driven insights on restaurant performance, pricing, and market segments

06

Case Study 5 — Bank Marketing Campaign Analysis

Property	Details
Dataset	Bank Marketing Dataset (UCI / Kaggle)
Key Columns	Age, job, marital, education, balance, housing, loan, campaign, poutcome, y (subscribed: yes/no)
Business Question	Which customer profiles are most likely to subscribe to a term deposit?
Modules Applied In	Python — NumPy, Pandas, Matplotlib, Seaborn (Modules 6–10)
Expected Output	Full EDA report with visualisations identifying high-conversion customer segments

PRO TIP — Download All Datasets Today

Go to [Kaggle.com](https://www.kaggle.com) and download all five datasets right now. Create a folder called 'DA Program Datasets' on your laptop. Having the data on hand means you can explore ahead of schedule — and the best students always do.

TRY THIS LIVE

Open OnlineRetail.csv in Excel. Using the five EDA questions from Session 5, spend 10 minutes answering: How many rows and columns? What does each column mean? Are there missing CustomerID values? What is the date range? What business question does this data help answer? Write your answers — we will compare across the batch.

ANALYSTS WITH PORTFOLIO PROJECTS GET HIRED FASTER

3x

Source: LinkedIn Talent Insights India 2023 — data analysts with project portfolios receive 3x more interview callbacks than those with only certifications

CONCEPT — Your Portfolio Starts Now

Every case study you complete in this program is a portfolio project. Save your Excel files, SQL queries, Python notebooks, and Power BI reports. By the end of this course, you will have five complete, real-world analytics projects to show any recruiter. This is your competitive advantage over candidates who only have certificates.

“The best analysts are not the ones who know the most tools — they are the ones who have solved the most real problems.”

WHERE TO GO NEXT

Module 1 is complete. You now know what analytics is, where data comes from, how to define KPIs, how a project is structured, and how to approach a new dataset. Module 2 builds the statistical foundation that makes your analysis rigorous.

Explore deeper — Browse the five Kaggle datasets you downloaded. Read the 'Data Card' (dataset description) for each one — understanding what the data collector intended tells you a lot about how to use it.

Try on your own — Write a one-paragraph 'case brief' for any one of the five case studies — state the business context, the problem, and what you think the analysis should recommend. We will revisit this at the end of the module.

Connect the dots — Module 2 (Statistics and Probability) begins with descriptive statistics — the foundation of every EDA you will run on these datasets.

Going further — Read 'Storytelling with Data' by Cole Nussbaumer Knaflic — the best book on turning analytics outputs into business presentations that actually drive decisions.

AI PROMPT — ASK CHATGPT / GEMINI / CLAUDE

Act as a data analytics mentor. I am a fresher starting my first analytics project on the Telco Customer Churn dataset. Guide me through the first steps — what questions should I ask about the business, what should I look for in the data, and what tools should I use at each stage? Keep it practical and under 200 words in simple English.

REVIEW



INTERVIEW PREP · MODULE 1

Interview Questions

◆ Introduction to Data Analytics

These questions test your understanding of core data analytics concepts. Rehearse answering out loud, aiming for 60-second responses. The hints are nudges to guide your thinking—not the answers. Use the model answers to compare your response after you've tried.

1

What is data analytics, and why is it important for organizations today?

MODEL ANSWER

Data analytics is the process of examining raw data to uncover patterns, draw conclusions, and generate insights that inform business decisions. It's important because organizations face massive data volumes daily; analytics transforms this data into actionable intelligence, enabling competitive advantage, cost optimization, risk reduction, and better customer understanding. In a data-driven world, organizations that analyze their data effectively outperform those that don't.

2

Explain the difference between data, information, and insights using a simple example.

Think about how raw numbers become meaningful and then actionable.

MODEL ANSWER

Data is raw, unprocessed facts—like a spreadsheet of transaction amounts. Information is data that has been organized, cleaned, and given context—like total monthly revenue grouped by product category. Insights are actionable conclusions drawn from information—like 'Product A has 40% higher margins and should get 60% of marketing budget.' Each builds on the previous level of understanding.

3

What are the four main types of analytics, and what is the purpose of each?

Recall: Descriptive, Diagnostic, Predictive, Prescriptive.

MODEL ANSWER

Descriptive analytics answers 'What happened?' by summarizing historical data—like monthly sales totals. Diagnostic analytics answers 'Why did it happen?' by identifying root causes—like analyzing why Q3 sales dropped. Predictive analytics answers 'What will happen?' using models to forecast trends—like predicting customer churn. Prescriptive analytics answers 'What should we do?' by recommending actions—like suggesting optimal pricing strategies to maximize revenue.

4

What is the role of a Data Analyst (DA) compared to a Data Engineer (DE)?

Focus on who prepares data and who analyzes it.

MODEL ANSWER

A Data Engineer builds and maintains the infrastructure, pipelines, and systems that collect, store, and process data efficiently. A Data Analyst uses this prepared data to explore, analyze, and generate insights that answer business questions. Engineers focus on 'How do we handle data at scale?' while Analysts focus on 'What does this data tell us?' Both roles are essential—engineering makes analysis possible.

5

Name at least three industry tools used for data analytics and mention one key use for each.

MODEL ANSWER

Excel is used for quick data exploration, calculations, and pivot tables. SQL is used for querying large databases efficiently and aggregating data from multiple tables. Python is used for advanced data manipulation, statistical analysis, and machine learning. Power BI is used for building interactive dashboards and visualizations that stakeholders can explore.

6

If you have a sales dataset, how might it look different in Excel, SQL, Python, and Power BI?

Think about the interface and capabilities of each tool.

MODEL ANSWER

In Excel, it's a rectangular grid where you can manually perform calculations and create simple charts. In SQL, it's queried from a database using SELECT statements, returning filtered or aggregated results. In Python, it's loaded as a DataFrame where you can programmatically transform, analyze, and visualize it. In Power BI, it's connected as a data source that powers interactive dashboards with slicers and drill-down capabilities.

7

What is the difference between structured and unstructured data? Give an example of each.

Consider tables vs. free-form text.

MODEL ANSWER

Structured data has a defined schema and is organized in tables with rows and columns—like a customer database with fields for Name, Age, Email, and Purchase Amount. Unstructured data lacks a predefined format—like customer reviews, social media posts, images, or audio files. Structured data is easier to query and analyze; unstructured data requires text mining or NLP techniques to extract meaning.

8

List three common data formats you might encounter in analytics work.

Think about files you open in Excel or import into Python.

MODEL ANSWER

CSV (Comma-Separated Values) is a plain text format widely used for exporting and importing data. JSON (JavaScript Object Notation) is a nested, hierarchical format often used for APIs and unstructured data. Excel (.xlsx) is a spreadsheet format that stores data in cells with formatting and formulas. SQL databases store structured data in normalized tables. Parquet and Apache formats are also used for big data.

9

What is a relational database management system (RDBMS), and why is it used?

MODEL ANSWER

An RDBMS organizes data into related tables using primary and foreign keys, allowing efficient storage and retrieval without redundancy. It enforces data integrity through constraints, supports complex queries with JOINS, and scales to millions of records. RDBMS systems like MySQL and PostgreSQL are industry standards because they reliably handle structured data, enable multi-user access, and provide robust backup and security.

10**Explain the difference between ETL and ELT processes in data handling.***Focus on when transformation happens relative to loading data.***MODEL ANSWER**

ETL (Extract, Transform, Load) pulls data from sources, cleans and transforms it before loading into a target system. ELT (Extract, Load, Transform) loads raw data first, then transforms it in the destination system. ETL is traditional and good for ensuring quality before storage; ELT leverages cloud computing power to transform large datasets after loading, often more cost-effective and flexible.

11**How does OLTP differ from OLAP in terms of use case?***Think about transactions vs. analysis.***MODEL ANSWER**

OLTP (Online Transaction Processing) is optimized for fast, frequent small operations—like a point-of-sale system recording individual sales. OLAP (Online Analytical Processing) is optimized for complex queries across large datasets—like analyzing sales by region, product, and time period. OLTP prioritizes speed of individual transactions; OLAP prioritizes speed of aggregated queries.

12**Why is domain knowledge important for a data analyst working on a retail project?***Consider how business context affects which metrics matter.***MODEL ANSWER**

Domain knowledge helps analysts understand which metrics matter (e.g., AOV, conversion rate, churn) and interpret findings in business context. Without it, analysts might miss insights or recommend irrelevant metrics. A strong domain understanding allows analysts to ask smarter questions, validate results for reasonableness, and communicate findings in terms stakeholders care about—like profit margin or customer lifetime value.

13**Give an example of a key performance indicator (KPI) for the finance domain.****MODEL ANSWER**

Return on Investment (ROI) is a common financial KPI that measures profit relative to investment. Other examples include Debt-to-Equity Ratio, Net Profit Margin, or Liquidity Ratio. KPIs track whether a business is meeting strategic goals. In finance, KPIs typically focus on profitability, risk, and operational efficiency.

14**How would you define and use the metric 'Average Order Value' (AOV) in an analytics project?***Think about what AOV tells you about customer behavior.***MODEL ANSWER**

AOV is calculated as Total Revenue divided by Number of Orders. It measures the average amount spent per transaction and reveals customer spending patterns. A rising AOV suggests successful upselling; declining AOV might indicate price sensitivity or fewer high-value customers. Analysts use AOV to track revenue health, compare segments, and identify opportunities to increase transaction value.

15

If a client from the healthcare sector asks for a dashboard, what kind of business metrics might you include?

Metrics could relate to patients, costs, or outcomes.

MODEL ANSWER

Healthcare dashboards might track patient volume (admissions, readmissions), clinical outcomes (mortality rate, infection rates), operational metrics (bed utilization, average length of stay), and financial metrics (revenue per patient, cost per treatment). The choice depends on the client's priorities—a hospital focused on quality tracks outcomes; one focused on efficiency tracks utilization and cost per case.

16

What are the main steps in a typical data analytics workflow?

Start from understanding the problem and end at sharing insights.

MODEL ANSWER

The workflow is: (1) Requirement Gathering—understand the business question, (2) Data Collection—gather relevant data, (3) Cleaning—remove errors and inconsistencies, (4) Transformation—prepare data for analysis, (5) Modeling—apply statistical or analytical techniques, (6) Reporting—create visualizations and summaries, (7) Insights & Presentation—communicate findings to stakeholders for decision-making.

17

Explain why data cleaning is a critical step before analysis.

Consider what happens if there are errors or missing values.

MODEL ANSWER

Data cleaning removes errors, inconsistencies, duplicates, and missing values that distort analysis. Dirty data leads to wrong conclusions and poor decisions. For example, if a price column has typos or blank cells, any revenue calculation is invalid. Analysts often spend 60-70% of project time on cleaning because data quality directly determines analysis quality—'garbage in, garbage out.'

18

Describe the difference between data collection and data transformation in the analytics process.

One is about gathering, the other about changing format or structure.

MODEL ANSWER

Data collection is the process of gathering raw data from various sources—databases, APIs, files, surveys. Data transformation is the process of converting collected data into a structured, clean format suitable for analysis—combining tables, calculating new columns, standardizing formats, or aggregating rows. Collection creates the raw material; transformation prepares it for use.

19

How would you approach requirement gathering for a new analytics project?

Think about communication with stakeholders.

MODEL ANSWER

Effective requirement gathering involves (1) Meeting stakeholders to understand their business goals, (2) Clarifying success metrics and KPIs, (3) Identifying data sources and limitations, (4) Asking probing questions about edge cases and data quality, (5) Documenting assumptions and constraints, (6) Setting realistic timelines. Good requirements prevent scope creep and misalignment.

20

Imagine you have completed your analysis. What are some best practices for presenting insights to a non-technical audience?

Consider the use of visuals and simple language.

MODEL ANSWER

Use clear, simple language avoiding jargon. Focus on business impact, not technical details. Use visuals—charts, dashboards, infographics—instead of numbers alone. Lead with conclusions, then provide supporting evidence. Tell a story—context, findings, recommendations. Tailor the message to the audience's interests and priorities. Invite questions and be ready to explain your methodology.

21

What is Exploratory Data Analysis (EDA), and why is it important at the start of a project?

MODEL ANSWER

EDA is the process of visually and statistically exploring data to understand its structure, patterns, and anomalies before formal analysis. It answers initial questions like: What's the data distribution? Are there missing values? What are the relationships between variables? EDA is essential because it guides the analytical direction, reveals data quality issues early, and prevents wasted time on flawed hypotheses.

22

Describe the difference between univariate and bivariate analysis with examples.

Think about analyzing one variable vs. two.

MODEL ANSWER

Univariate analysis examines one variable at a time—calculating mean, median, distribution, or variance. Example: 'What's the average customer age?' Bivariate analysis examines the relationship between two variables—using correlation, regression, or crosstabs. Example: 'How does customer age relate to purchase frequency?' Univariate is descriptive; bivariate reveals dependencies and associations.

23

How can you identify outliers in a dataset during EDA?

Consider statistical methods or visual tools.

MODEL ANSWER

Visually, box plots and scatter plots highlight extreme values. Statistically, the IQR (Interquartile Range) method flags values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$. Z-score method flags values more than 3 standard deviations from the mean. Context matters—an outlier might be valid (a high-value customer) or an error (a typo). Always investigate before removing outliers.

24

If you are given a dataset in Excel, what is one simple way to spot trends or relationships between two columns?

Think about charts or correlation functions.

MODEL ANSWER

Create a scatter plot with one variable on the x-axis and another on the y-axis—visual patterns immediately show trends, clusters, or correlations. Alternatively, use Excel's CORREL function to calculate the correlation coefficient, showing strength and direction of relationship. Conditional formatting with color gradients also highlights patterns. These approaches are quick and don't require coding.

25

What is a multivariate analysis, and when would you use it?

Think about analyzing more than two variables together.

MODEL ANSWER

Multivariate analysis examines relationships among three or more variables simultaneously. Examples include multiple regression (predicting sales from price, marketing spend, and seasonality) or principal component analysis (reducing many variables into key dimensions). Use multivariate analysis when you suspect multiple factors influence an outcome and want to understand their combined effect and relative importance.

26

Suppose you notice an unexpected spike in sales in your EDA. What steps would you take to investigate this further?

Think about checking data quality and external events.

MODEL ANSWER

First, verify the spike is real by checking for data quality issues—duplicates, errors, or typos. Second, investigate timing—did a promotion, holiday, or competitor action coincide with the spike? Third, segment the data—did the spike occur in one region, product, or customer segment? Fourth, compare to previous years for seasonality. Finally, validate with stakeholders before drawing conclusions. Spikes often result from external factors, not true demand changes.

27

Why is it important to understand the flow of a dataset before starting your analysis?

Consider how data moves from source to report.

MODEL ANSWER

Understanding data flow—how it moves from source systems through ETL pipelines to analytics databases—reveals where errors can occur and helps validate data freshness and completeness. It clarifies data lineage (which original fields contributed to calculated columns), identifies dependencies between tables, and explains delays or inconsistencies. Analysts who understand flow ask better questions and spot data issues faster.

28

In a case study, how do you turn a business question into an analytics problem?

Think about translating goals into measurable metrics.

MODEL ANSWER

Start by clarifying the business goal—e.g., 'Increase revenue' is vague. Reframe as an analytics problem: 'Which product categories have declining year-over-year revenue, and what are the customer behavior changes driving the decline?' Define measurable success metrics (growth rate, customer retention, AOV). Identify available data sources and timeframes. Break the question into sub-questions that guide analysis. This translation ensures the analytics work directly supports decision-making.

29

Imagine you are presented with a new case study dataset. What initial checks would you perform before analysis?

Look for missing values, formats, and consistency.

MODEL ANSWER

First, load the data and inspect structure—row count, column names, data types. Check for missing values and understand their distribution. Verify data formats are consistent (e.g., dates aren't mixed text and datetime). Identify duplicates. Look for obvious errors—negative values where impossible, impossible date ranges. Validate sample records against source documentation. Review basic statistics (min, max, mean) for outliers. Only after these checks begin analysis.

30

How would you use Excel to quickly summarize a dataset and spot any anomalies?

Consider pivot tables or conditional formatting.

MODEL ANSWER

Use a Pivot Table to aggregate data by dimensions (region, product, time period) and spot unusual patterns—like one region with zero sales. Apply Conditional Formatting with color scales to highlight high/low values visually. Use the SUBTOTAL function to summarize filtered subsets. Create a summary statistics sheet with min, max, mean, and std dev for each numeric column. These Excel-native tools are fast and require no coding.

31

How can you use SQL to filter data for only the last month's transactions in a sales table?

MODEL ANSWER

Use a WHERE clause with a date function: `SELECT * FROM sales WHERE MONTH(transaction_date) = MONTH(CURRENT_DATE) AND YEAR(transaction_date) = YEAR(CURRENT_DATE);` OR `SELECT * FROM sales WHERE transaction_date >= DATE_TRUNC('month', CURRENT_DATE);` The exact syntax depends on the SQL database (MySQL, PostgreSQL, SQL Server). The key is comparing transaction_date to the current date.

32

Give an example of how Python can be used for basic data cleaning.

Consider removing duplicates or filling missing values.

MODEL ANSWER

Import pandas and load your CSV: `df = pd.read_csv('data.csv')`. Remove duplicates: `df.drop_duplicates()`. Handle missing values: `df.fillna(df.mean())` or `df.dropna()`. Clean text columns: `df['name'] = df['name'].str.strip().str.lower()`. Convert data types: `df['date'] = pd.to_datetime(df['date'])`. Export cleaned data: `df.to_csv('clean_data.csv')`. Python handles these tasks much faster than manual Excel operations on large datasets.

33

If you want to visualize sales by region in Power BI, what kind of chart would you choose and why?

Consider which chart best shows comparisons across categories.

MODEL ANSWER

A bar chart (horizontal or vertical) is ideal for comparing sales values across regions—it makes differences immediately obvious. Alternatively, a column chart works well, especially if you have few regions. If you want to show both total sales and regional share, a pie or donut chart works but is less precise. For trends over time by region, use a line chart with different colors per region. Choose based on your audience and the key insight.

34

Suppose your data is in JSON format. How would you load and work with this data in Python?

Think about libraries or functions that handle JSON.

MODEL ANSWER

Use the json module for simple JSON files: `import json; data = json.load(open('file.json'))`; For JSON as pandas DataFrame: `import pandas as pd; df = pd.read_json('file.json')`; Extract nested fields using pandas indexing: `df['nested_field'] = df['parent'].apply(lambda x: x['child'])`. For complex nested JSON, normalize it first: `df = pd.json_normalize(data)`. Once in DataFrame format, use standard pandas operations for analysis.

35

How does understanding both business metrics and technical data help you deliver better analytics solutions?

Consider how analysis is used for decision-making.

MODEL ANSWER

Business understanding ensures you analyze the right metrics and frame findings in terms stakeholders understand—like profit impact rather than statistical significance. Technical skill ensures your data is clean, your methods are sound, and your code is reproducible. Combined, they enable you to translate vague business problems into precise analytical solutions and deliver insights stakeholders will act on. Without business sense, perfect analysis gets ignored.

36

If a dataset has both structured tables and unstructured text reviews, how might your approach to analysis change?

Think about tools and methods suitable for each data type.

MODEL ANSWER

For structured data (tables), use SQL queries, pivot tables, and traditional statistical analysis. For unstructured text reviews, use text mining and NLP—tokenization, sentiment analysis, keyword extraction, topic modeling. You might combine insights: correlate structured metrics (rating, purchase amount) with text sentiment to understand what drives satisfaction. Different data types require different techniques; a comprehensive analysis often integrates both.

37

You are asked to automate a monthly report that combines data from Excel and SQL. Briefly describe how you would approach this task.

Consider data extraction, transformation, and automation tools.

MODEL ANSWER

Extract data from SQL using Python (`pd.read_sql`) and from Excel using `pd.read_excel()`. Combine and transform the datasets—clean, calculate KPIs, aggregate. Build report structure—create charts and tables in a new Excel file or HTML dashboard. Automate using a scheduler: Windows Task Scheduler or cron job to run a Python script monthly. Alternative: use tools like n8n or Zapier for low-code automation, or schedule scripts in cloud environments like AWS Lambda or Google Cloud Functions.

38

How could you use ChatGPT or another AI tool to help you understand a complex dataset or troubleshoot a data cleaning issue?

Think about asking for code samples or explanations, but also how you would verify the AI's suggestions.

MODEL ANSWER

Ask ChatGPT to explain a statistical concept or suggest Python code for common tasks—outlier removal, missing value imputation, data transformation. Use it to brainstorm analysis approaches or troubleshoot errors. However, always verify AI suggestions: test code on sample data, validate logic, check for edge cases, and compare against documentation. Never blindly trust AI output—it can make confident-sounding mistakes. Use AI as a learning partner, not a replacement for critical thinking.

39

What are some risks of relying solely on AI-generated insights or code when working on a data analytics project?

Consider data privacy, accuracy, and the importance of human judgment.

MODEL ANSWER

AI can hallucinate incorrect conclusions or miss important nuances about business context. AI-generated code might work on sample data but fail at scale or with edge cases. Privacy risks arise if you upload sensitive data to public AI services. AI insights lack the business judgment to distinguish correlation from causation or understand implications. AI cannot replace domain expertise. Best practice: use AI as a tool to accelerate work, but always review, validate, and apply human judgment before sharing insights or decisions.